

Name
Date
Course

Polynomials (Part II): Long Division and Synthetic Division

Review

Perform the following division.

① $(5x^3 + 5 - 20x) \div (-2 + 3x + x^2)$

$$\begin{array}{r} 5x-15 \\ x^2+3x-2 \overline{) 5x^3 -20x+5} \\ \underline{5x^3+15x^2-10x} \\ -15x^2-10x+5 \\ \underline{-15x^2-45x+30} \\ 55x-25 \end{array} \quad \boxed{5x-15 + \frac{55x-25}{x^2+3x-2}}$$

② $(6x^5 - 9x^6 + 8) \div (3x^3 + 3)$

$$\begin{array}{r} -3x^3+2x^2+3 \\ 3x^3+3 \overline{) -9x^6+6x^5 +8} \\ \underline{-9x^6 -9x^3} \\ 6x^5+9x^3+8 \\ \underline{6x^5 +6x^2} \\ 9x^3-6x^2+8 \\ \underline{9x^3 +9} \\ -6x^2-1 \end{array} \quad \begin{array}{l} -3x^3+2x^2+3 + \frac{-6x^2-1}{3x^3+3} \\ \boxed{-3x^3+2x^2+3 - \frac{6x^2+1}{3x^3+3}} \end{array}$$

③ $(15 + 10x + 3x^2 - 2x^4) \div (x+1)$

$$\begin{array}{r} -2x^3+2x^2+x+9 \\ x+1 \overline{) -2x^4 +10x+15} \\ \underline{-2x^4-2x^3} \\ 2x^3+3x^2+10x+15 \\ \underline{2x^3+2x^2} \\ x^2+10x+15 \\ \underline{x^2+x} \\ 9x+15 \\ \underline{9x+9} \\ 6 \end{array} \quad \boxed{-2x^3+2x^2+x+9 + \frac{6}{x+1}}$$

④ $(10 - x^5 - 8x^2) \div (-5 + x^2)$

$$\begin{array}{r} -x^3-5x-8 \\ x^2-5 \overline{) -x^5 +10} \\ \underline{-x^5+5x^3} \\ -5x^3-8x^2+10 \\ \underline{-5x^3 +25x} \\ -8x^2-25x+10 \\ \underline{-8x^2 +40} \\ -25x-30 \end{array} \quad \begin{array}{l} -x^3-5x-8 + \frac{-25x-30}{x^2-5} \\ \boxed{-x^3-5x-8 - \frac{25x+30}{x^2-5}} \end{array}$$

More Long Division Problems

⑤ $(-4x^3 + x^2 + 4x) \div (x-2)$

$$\begin{array}{r}
 -4x^2 - 7x - 10 \\
 x-2 \overline{) -4x^3 + x^2 + 4x} \\
 \underline{-4x^3 + 8x^2} \\
 -7x^2 + 4x \\
 \underline{-7x^2 + 14x} \\
 -10x \\
 \underline{-10x + 20} \\
 -20
 \end{array}
 \quad -4x^2 - 7x - 10 - \frac{20}{x-2}$$

⑥ $(x^3 - 4x^2 + 5x) \div (x^2 + 4)$

$$\begin{array}{r}
 x - 4 \\
 x^2 + 4 \overline{) x^3 - 4x^2 + 5x} \\
 \underline{x^3 + 4x} \\
 -4x^2 + x \\
 \underline{-4x^2 - 16} \\
 x + 16
 \end{array}
 \quad x - 4 + \frac{x + 16}{x^2 + 4}$$

⑦ $(x^3 + 6) \div (x-4)$

$$\begin{array}{r}
 x^2 + 4x + 16 \\
 x-4 \overline{) x^3 + 6} \\
 \underline{x^3 - 4x^2} \\
 4x^2 + 6 \\
 \underline{4x^2 - 16x} \\
 16x + 6 \\
 \underline{16x - 64} \\
 70
 \end{array}
 \quad x^2 + 4x + 16 + \frac{70}{x-4}$$

⑧ $(4x^4 - 35) \div (x+1)$

$$\begin{array}{r}
 4x^3 - 4x^2 + 4x - 4 \\
 x+1 \overline{) 4x^4 - 35} \\
 \underline{4x^4 + 4x^3} \\
 -4x^3 - 35 \\
 \underline{-4x^3 - 4x^2} \\
 4x^2 - 35 \\
 \underline{4x^2 + 4x} \\
 -4x - 35 \\
 \underline{-4x - 4} \\
 -31
 \end{array}
 \quad 4x^3 - 4x^2 + 4x - 4 - \frac{31}{x+1}$$

⑨ $(-2x^3) \div (x+5)$

$$\begin{array}{r}
 -2x^3 + 10x - 50 \\
 x+5 \overline{) -2x^3} \\
 \underline{-2x^3 - 10x^2} \\
 10x^2 \\
 \underline{10x^2 + 50x} \\
 -50x \\
 \underline{-50x - 250} \\
 250
 \end{array}
 \quad -2x^3 + 10x - 50 + \frac{250}{x+5}$$

⑩ $(x^4) \div (x-3)$

$$\begin{array}{r}
 x^3 + 3x^2 + 9x + 27 \\
 x-3 \overline{) x^4} \\
 \underline{x^4 - 3x^3} \\
 3x^3 \\
 \underline{3x^3 - 9x^2} \\
 9x^2 \\
 \underline{9x^2 - 27x} \\
 27x \\
 \underline{27x - 81} \\
 81
 \end{array}
 \quad x^3 + 3x^2 + 9x + 27 + \frac{81}{x-3}$$

Synthetic Division

Synthetic Division: An algorithm used to divide polynomials that eliminates the need to write variables.
 * This algorithm can only be used when dividing by expressions of the form $x+C$ or $x-C$, where C is a constant.

Example

$$(2x^4 + x^3 - 23x^2 - 46x - 24) \div (x-4)$$

Step I: Write the numbers as seen above, remembering to change the sign of the constant in the divisor.

Step II: Bring the first number from the dividend down.

Step III: The remaining part of the algorithm is 2-step cycles. 1. Multiply these two numbers and write the product. Add these two numbers and write the sum. Repeat this process.

Divide the following polynomials using synthetic division.

⑪ $(x^4 - 7x^3 + x^2 + 27x + 18) \div (x + 1)$

$$\begin{array}{r|rrrrr} -1 & 1 & -7 & 1 & 27 & 18 \\ & & -1 & 8 & -9 & -18 \\ \hline & 1 & -8 & 9 & 18 & 0 \end{array}$$

$$\boxed{x^3 - 8x^2 + 9x + 18}$$

⑫ $(3x^6 - 14x^5 - 16x^4 + 46x^3 + 49x^2 - 16x - 20) \div (x - 5)$

$$\begin{array}{r|rrrrrrrr} 5 & 3 & -14 & -16 & 46 & 49 & -16 & -20 \\ & & 15 & 5 & -55 & -45 & 20 & 20 \\ \hline & 3 & 1 & -11 & -9 & 4 & 4 & 0 \end{array}$$

$$\boxed{3x^5 + x^4 - 11x^3 - 9x^2 + 4x + 4}$$

⑬ $(-6x^4 + 2x^3 - 7x^2 + 4x - 8) \div (x^2 + 4x - 7)$

Cannot use synthetic division

⑭ $(-5x^3 - 32x^2 - 52x - 16) \div (x + 2)$

$$\begin{array}{r|rrrr} -2 & -5 & -32 & -52 & -16 \\ & & 10 & 44 & 16 \\ \hline & -5 & -22 & -8 & 0 \end{array}$$

$$\boxed{-5x^2 - 22x - 8}$$

⑮ $(6x^4 - 13x^3 - 28x^2 + 41x - 6) \div (x - 3)$

$$\begin{array}{r|rrrrr} 3 & 6 & -13 & -28 & 41 & -6 \\ & & 18 & 12 & -39 & 6 \\ \hline & 6 & 5 & -13 & 2 & 0 \end{array}$$

$$\boxed{6x^3 + 5x^2 - 13x + 2}$$

⑯ $(-x^4 - 3x^3 + 15x^2 - 17x + 6) \div (x + 6)$

$$\begin{array}{r|rrrrr} -6 & -1 & -3 & 15 & -17 & 6 \\ & & 6 & -18 & 18 & -6 \\ \hline & -1 & 3 & -3 & 1 & 0 \end{array}$$

$$\boxed{-x^3 + 3x^2 - 3x + 1}$$

$$(17) (4x^5 + 19x^4 - 45x^3 - 70x^2 + 116x - 24) \div (x-2)$$

$$\begin{array}{r|rrrrrr} 2 & 4 & 19 & -45 & -70 & 116 & -24 \\ & & 8 & 54 & 18 & -104 & 24 \\ \hline & 4 & 27 & 9 & -52 & 12 & 0 \end{array}$$

$$\boxed{4x^4 + 27x^3 + 9x^2 - 52x + 12}$$

$$(18) (9x^5 + x^4 + 10x^3 + 4x^2 - 3x + 1) \div (x^2 + 4)$$

Cannot use synthetic division

$$(19) (-12x^3 - 47x^2 + 10x + 24) \div (x+4)$$

$$\begin{array}{r|rrrr} -4 & -12 & -47 & 10 & 24 \\ & & 48 & -4 & -24 \\ \hline & -12 & 1 & 6 & 0 \end{array}$$

$$\boxed{-12x^2 + x + 6}$$

$$(20) (8x^3 - 30x^2 + 16x + 6) \div (x-1)$$

$$\begin{array}{r|rrrr} 1 & 8 & -30 & 16 & 6 \\ & & 8 & -22 & -6 \\ \hline & 8 & -22 & -6 & 0 \end{array}$$

$$\boxed{8x^2 - 22x - 6}$$

$$(21) (-25 + 13x^4 + 48x^2 - 2x^6) \div (x-3)$$

$$(-2x^6 + 13x^4 + 48x^2 - 25) \div (x-3)$$

$$\begin{array}{r|rrrrrrrr} 3 & -2 & 0 & 13 & 0 & 48 & 0 & -25 \\ & & -6 & -18 & -15 & -45 & 9 & 27 \\ \hline & -2 & -6 & -5 & -15 & 3 & 9 & 2 \end{array}$$

$$\boxed{-2x^5 - 6x^4 - 5x^3 - 15x^2 + 3x + 9 + \frac{2}{x-3}}$$

$$(22) (4x - 15 + x^5) \div (x+2)$$

$$(x^5 + 4x - 15) \div (x+2)$$

$$\begin{array}{r|rrrrrr} -2 & 1 & 0 & 0 & 0 & 4 & -15 \\ & & -2 & 4 & -8 & 16 & -40 \\ \hline & 1 & -2 & 4 & -8 & 20 & -55 \end{array}$$

$$\boxed{x^4 - 2x^3 + 4x^2 - 8x + 20 - \frac{55}{x+2}}$$

$$(23) (x^6 + 10 - 3x^3 - 5x) \div (x-2)$$

$$(x^6 - 3x^3 - 5x + 10) \div (x-2)$$

$$\begin{array}{r|rrrrrrr} 2 & 1 & 0 & 0 & -3 & 0 & -5 & 10 \\ & & 2 & 4 & 8 & 10 & 20 & 30 \\ \hline & 1 & 2 & 4 & 5 & 10 & 15 & 40 \end{array}$$

$$x^5 + 2x^4 + 4x^3 + 5x^2 + 10x + 15 + \frac{40}{x-2}$$

$$(24) (-10x^3 - 4x^4 + 27) \div (x+3)$$

$$(-4x^4 - 10x^3 + 27) \div (x+3)$$

$$\begin{array}{r|rrrrr} -3 & -4 & -10 & 0 & 0 & 27 \\ & & 12 & -6 & 18 & -54 \\ \hline & -4 & 2 & -6 & 18 & -27 \end{array}$$

$$-4x^3 + 2x^2 - 6x + 18 - \frac{27}{x+3}$$

$$(25) (x^3 + x^2 + x + 5) \div (3x-4)$$

Cannot use synthetic division

$$(26) (2x^4 - 8x^3 + 3x^2 - 2x) \div (x-4)$$

$$\begin{array}{r|rrrrr} 4 & 2 & -8 & 3 & -2 & 0 \\ & & 8 & 0 & 12 & 40 \\ \hline & 2 & 0 & 3 & 10 & 40 \end{array}$$

$$2x^3 + 3x + 10 + \frac{40}{x-4}$$

$$(27) (-8x^5 + 3x^4 + 11x^3 + 9x^2) \div (x+1)$$

$$\begin{array}{r|rrrrrr} -1 & -8 & 3 & 11 & 9 & 0 & 0 \\ & & 8 & -11 & 0 & -9 & 9 \\ \hline & -8 & 11 & 0 & 9 & -9 & 9 \end{array}$$

$$-8x^4 + 11x^3 + 9x - 9 + \frac{9}{x+1}$$

$$(28) (17x^5 - 20x + 100) \div (x^5 - x - 1)$$

Cannot use synthetic division

29) $(-x^4 + 1 - 3x) \div (2+x)$

$(-x^4 - 3x + 1) \div (x+2)$

$$\begin{array}{r|rrrrrr} -2 & -1 & 0 & 0 & -3 & 1 \\ & & 2 & -4 & 8 & -10 \\ \hline & -1 & 2 & -4 & 5 & -9 \end{array}$$

$-x^3 + 2x^2 - 4x + 5 - \frac{9}{x+2}$

If You may use a calculator.

30) $(3 + x - x^2 + 6x^5) \div (-3-x)$

$(6x^5 - x^2 + x + 3) \div (x-3)$

$$\begin{array}{r|rrrrrr} 3 & 6 & 0 & 0 & -1 & 1 & 3 \\ & & 18 & 54 & 162 & 483 & 1452 \\ \hline & 6 & 18 & 54 & 161 & 484 & 1455 \end{array}$$

$6x^4 + 18x^3 + 54x^2 + 161x + 484 + \frac{1455}{x-3}$

31) $(7 + 4x^2 - 10x^5) \div (8-x)$

Cannot use synthetic division.

32) $(x^3 + 10x - 1) \div (-x+4)$

Cannot use synthetic division